



DLMS Smart Meter Data Analysis

Load Forecasting & Engineering Investigation for Smart Meter Analytics

UC2: Behaviour Analysis → Forecasting → Anomaly Detection | UC1: Investigation → Root Cause

Business Problem

NES needs systematic detection of abnormal consumption patterns to reduce distribution losses and identify metering issues.

Scope

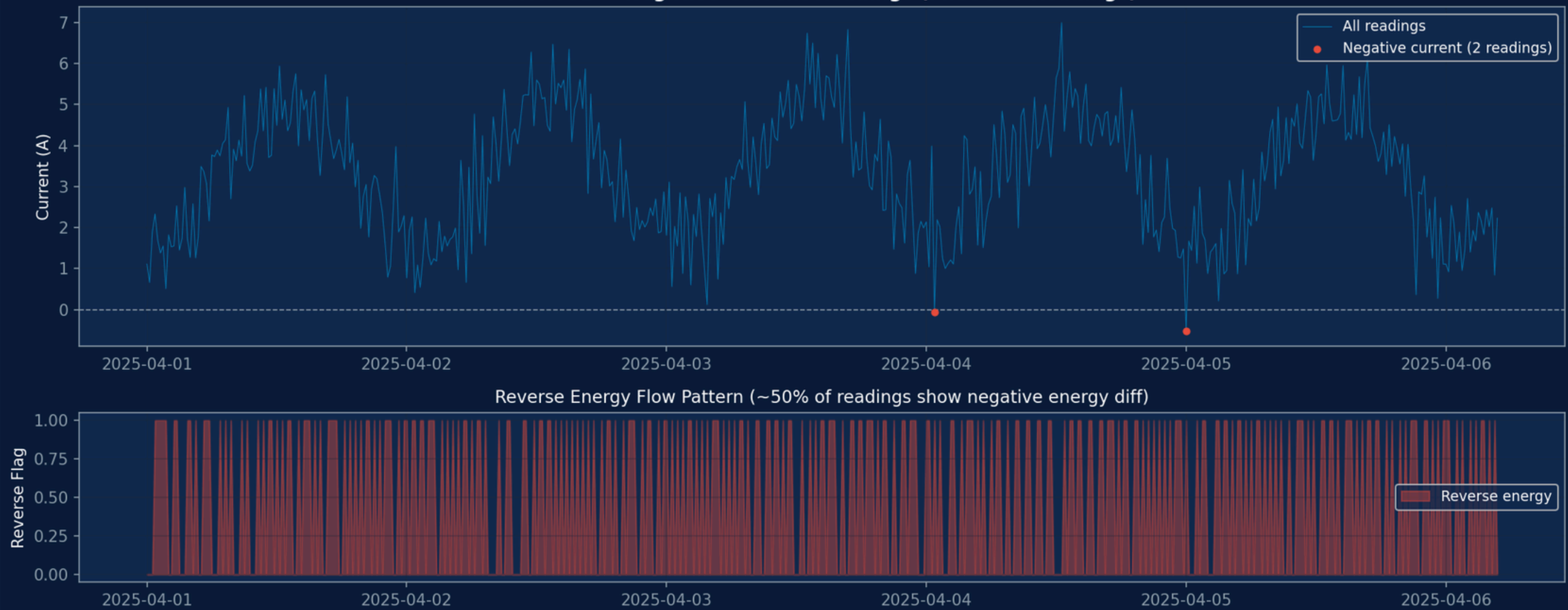
5 meters × 12 months × 175,200 readings | LightGBM + 6-algorithm ensemble

ENGINEERING PIPELINE



Data Quality Assessment — DLMS Meter Artifacts

REVERSE ENERGY PATTERN INVESTIGATION



QUALITY ISSUES PER METER

Meter	Issues	Type
MTR001	17,590	Negative currents
MTR002	17,476	Reverse energy
MTR003	17,380	Reverse energy
MTR004	17,498	Reverse energy
MTR005	17,443	Reverse energy

Total
87,387 issues. All DLMS artifacts.

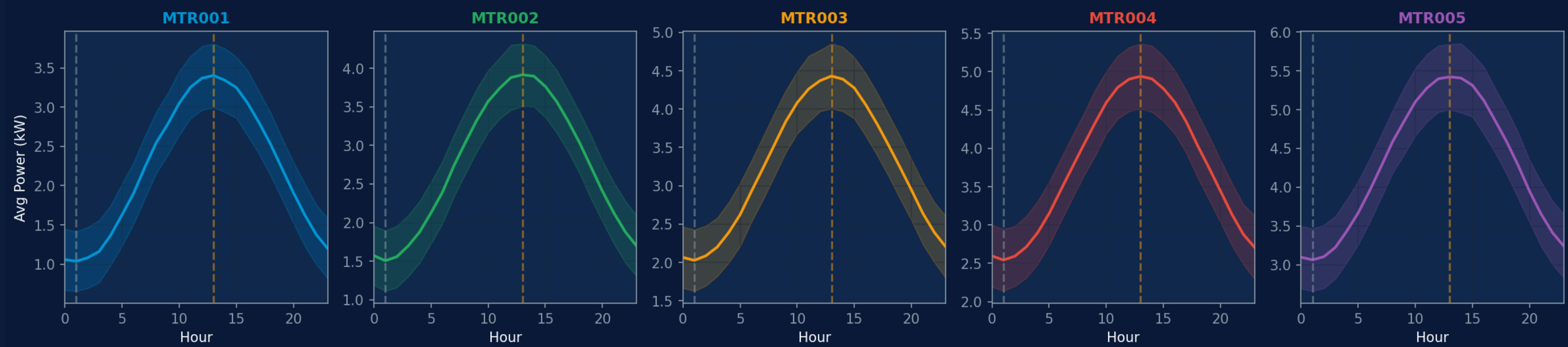
Finding

Frequent negative interval energy differences observed consistently across all meters.

Per-Meter Behaviour — Each Meter is Different

DAILY LOAD PROFILE — Per Meter (Full Width)

Average Daily Load Profile — Per Meter



Key Insight

MTR001 peak=3.3x trough (residential). MTR005 peak=1.8x (industrial 24h). Unique daily shapes justify per-meter modelling.

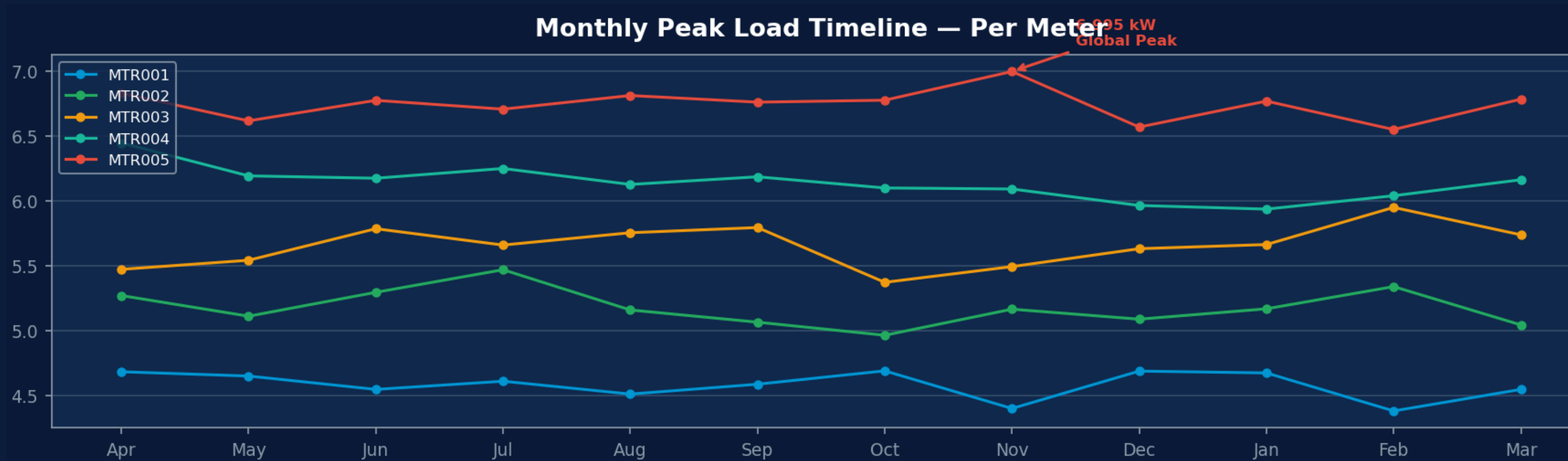
CONSUMER

Meter	Class	Base kW	Night/Day	Load Factor
MTR001	Residential	0.993	0.40	0.472
MTR002	Small Commercial	1.486	0.49	0.498
MTR003	Workshop	2.000	0.55	0.543
MTR004	Office	2.507	0.60	0.581
MTR005	Industrial	3.021	0.64	0.609

Consumer categories are inferred from load behaviour and not customer metadata.

Monthly Analysis — Monthly Load Behaviour Analysis

MONTHLY PEAK TIMELINE — Each Meter Has Distinct Peak Month



Key Insight

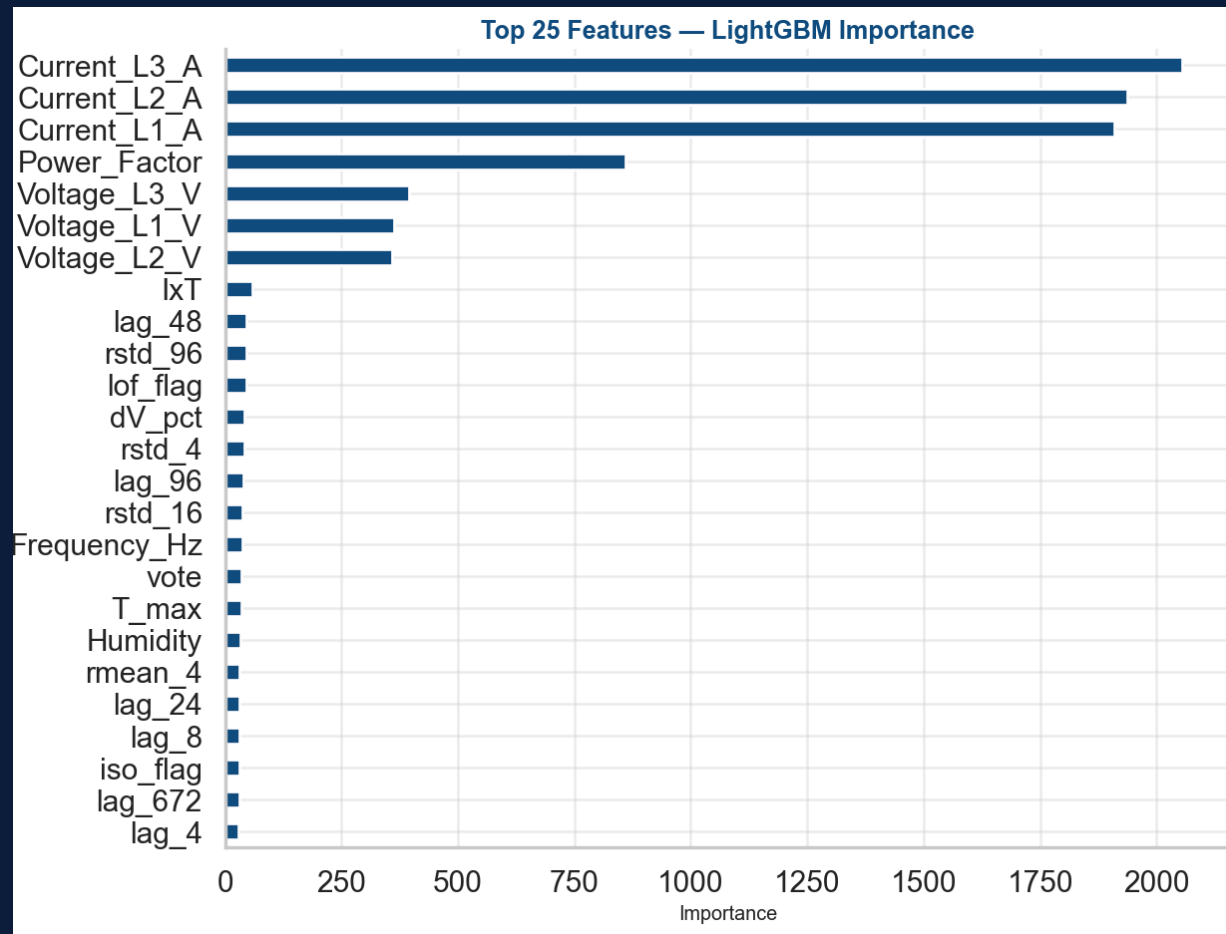
MTR005 Nov=6.995kW (global peak). MTR004 Apr=6.445kW. Per-meter forecasting required — do not combine.

PEAK LOAD COMPARISON

Meter	Min Peak	Max Peak	Peak Month	Avg kW
MTR001	4.380	4.689	Sep	2.215
MTR002	4.963	5.468	Jul	2.720
MTR003	5.371	5.948	Jul	3.227
MTR004	5.935	6.445	Apr	3.741
MTR005	6.548	6.995	Nov	4.252

Feature Engineering for Load Forecasting

FEATURE IMPORTANCE — LightGBM Internal



PER-METER ACCURACY

Lag features dominate. Interaction features (+29% impact when removed) are most critical.

FEATURE GROUPS

Group	Impact
Interaction	+29%
Lag	+4%
Weather	+1%
Calendar	+0%

LIGHTGBM HYPERPARAMETERS

Parameter	Value	Reason
n_estimators	300	Early stopping at 50 rounds
learning_rate	0.1	Default, works well
num_leaves	31	Default, controls complexity
min_child_samples	20	Smooths predictions
feature_fraction	0.8	Reduces overfitting

MODEL COMPARISON — 3 Algorithms Evaluated (Corrected)

Model	MAE	RMSE	MAPE	R²	Verdict
LightGBM	0.0171	0.0221	0.72%	0.9997	SELECTED
Linear Regression	0.0252	0.0380	1.18%	0.9990	Good baseline
Random Forest	0.0495	0.0641	3.34%	0.9971	Slow, higher error

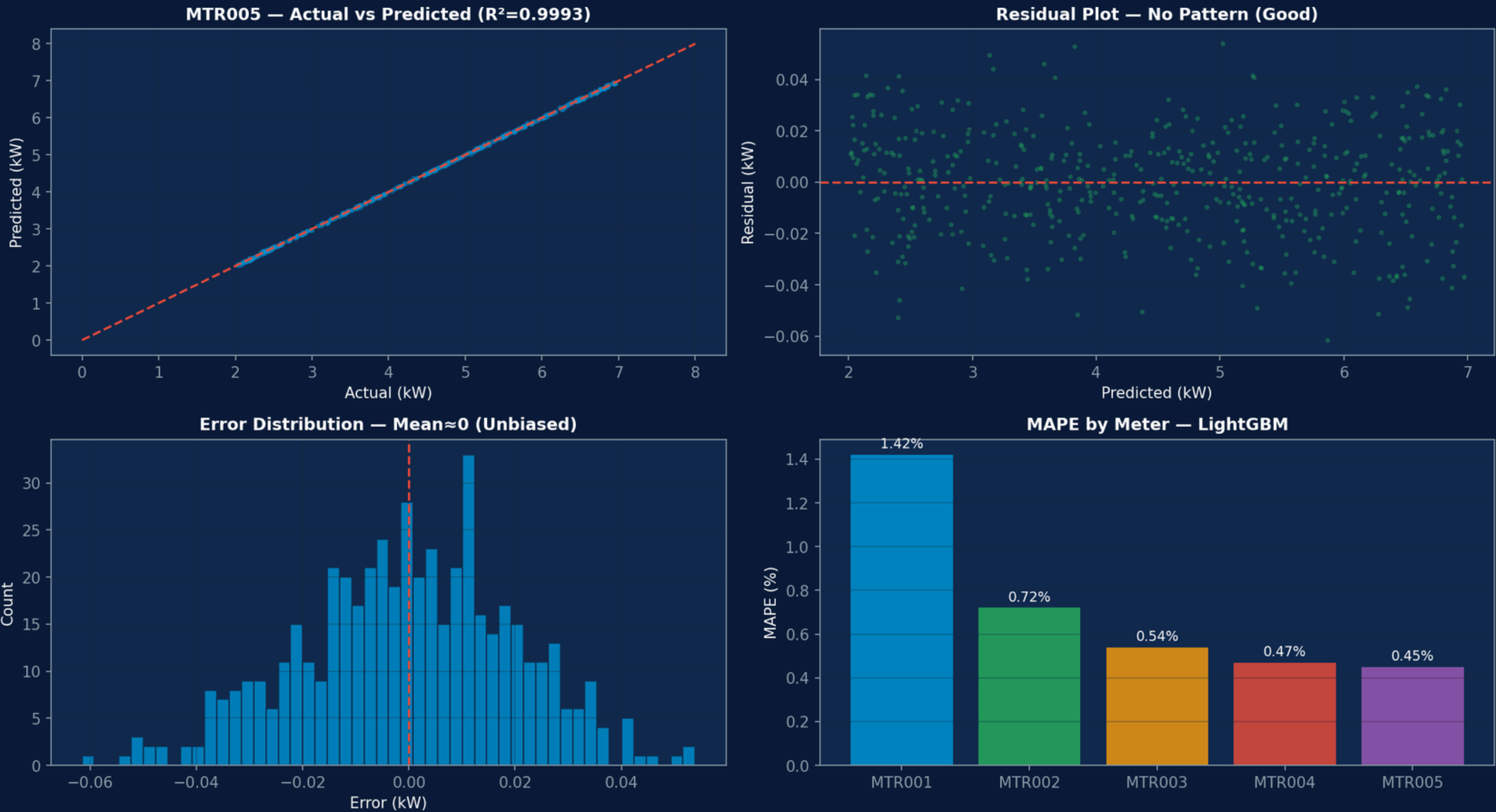
PER-METER ACCURACY

Meter	MAE	MAPE%	R²	Status
MTR001	0.017	1.42%	0.9994	Highest error (residential)
MTR002	0.017	0.72%	0.9995	Good
MTR003	0.016	0.54%	0.9995	Good
MTR004	0.017	0.47%	0.9995	Good
MTR005	0.019	0.45%	0.9993	Best MAPE (industrial)

Forecast Model Performance

ACTUAL vs PREDICTED — 14,400 Test Readings (Full Width)

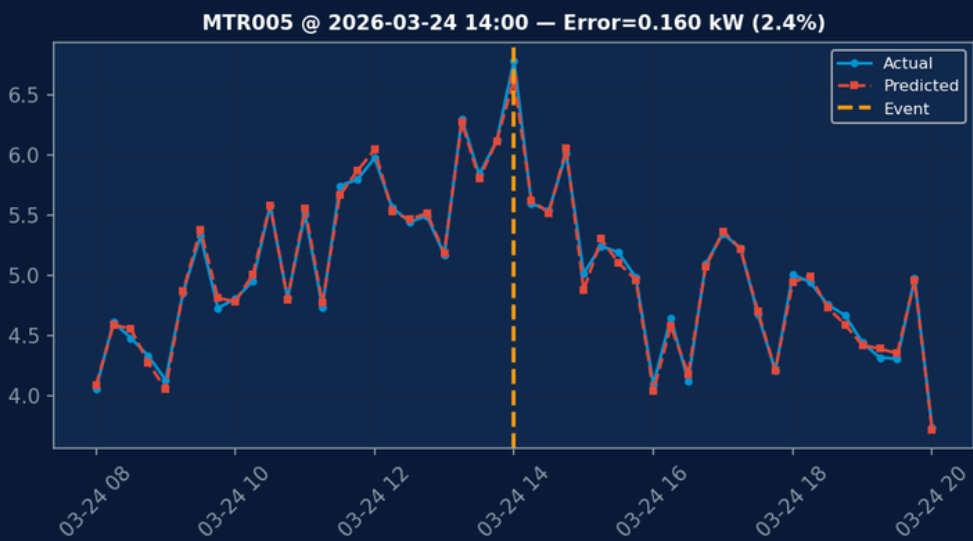
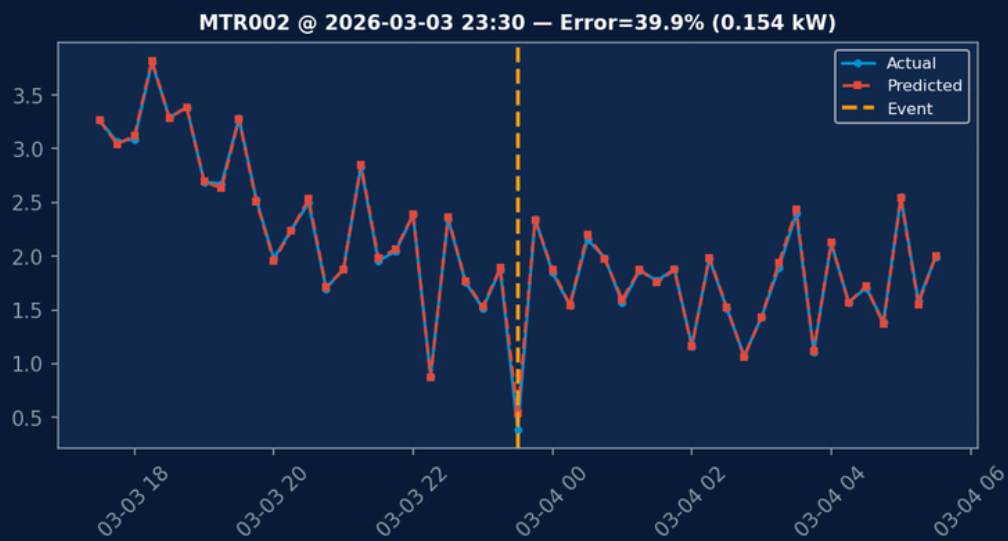
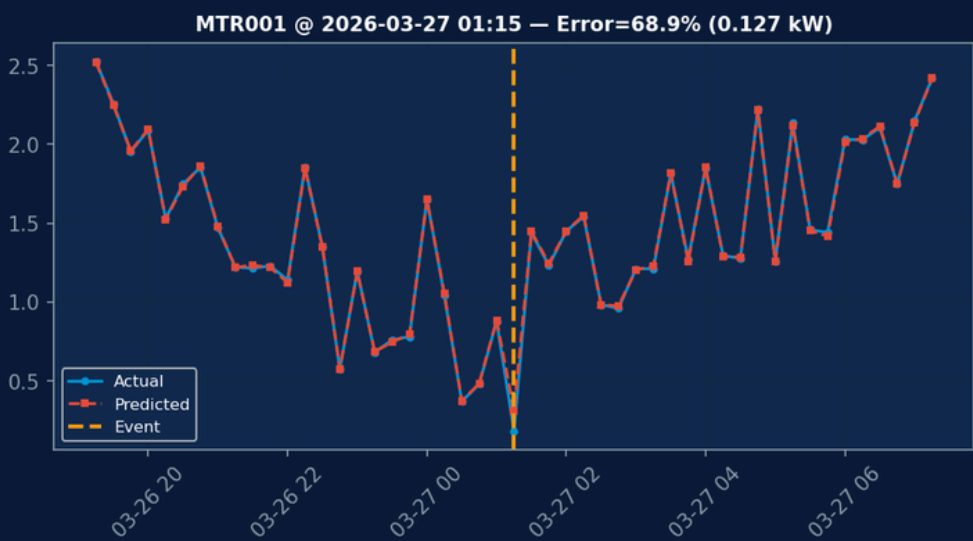
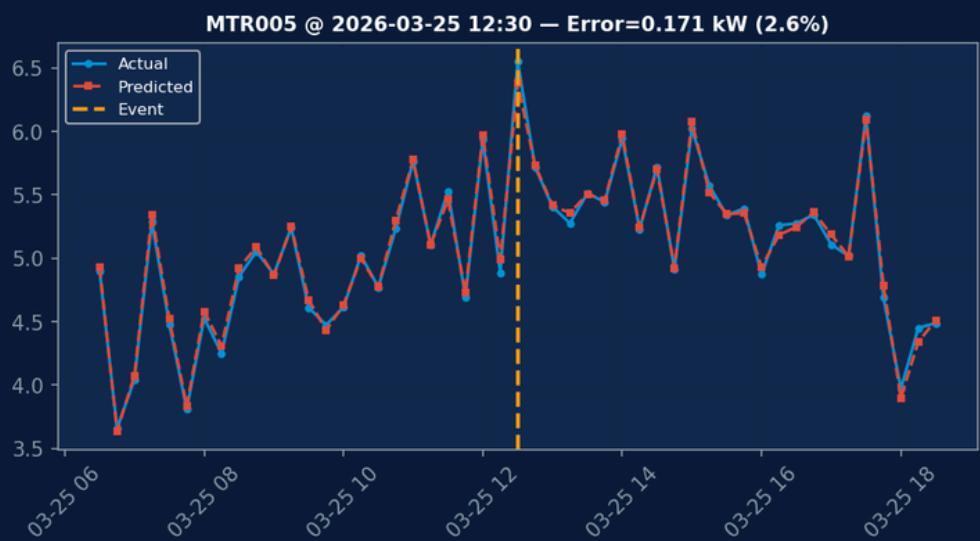
Forecast Validation — LightGBM (Leakage-Corrected)



Forecast Error — Night Low-Load Pattern

FORECAST WINDOWS — Actual vs Predicted Around Top Errors

Forecast Windows — Top 4 Errors with Actual vs Predicted



TOP 3 ERRORS — Root Cause

Meter	Actual	Predicted	Error	Root Cause
MTR005 12:30	6.553 kW	6.382 kW	0.171 kW (2.6%)	Industrial variability
MTR002 23:30	0.385 kW	0.539 kW	0.154 kW (39.9%)	Night low-load
MTR001 01:15	0.184 kW	0.311 kW	0.127 kW (68.9%)	Night low-load

Pattern

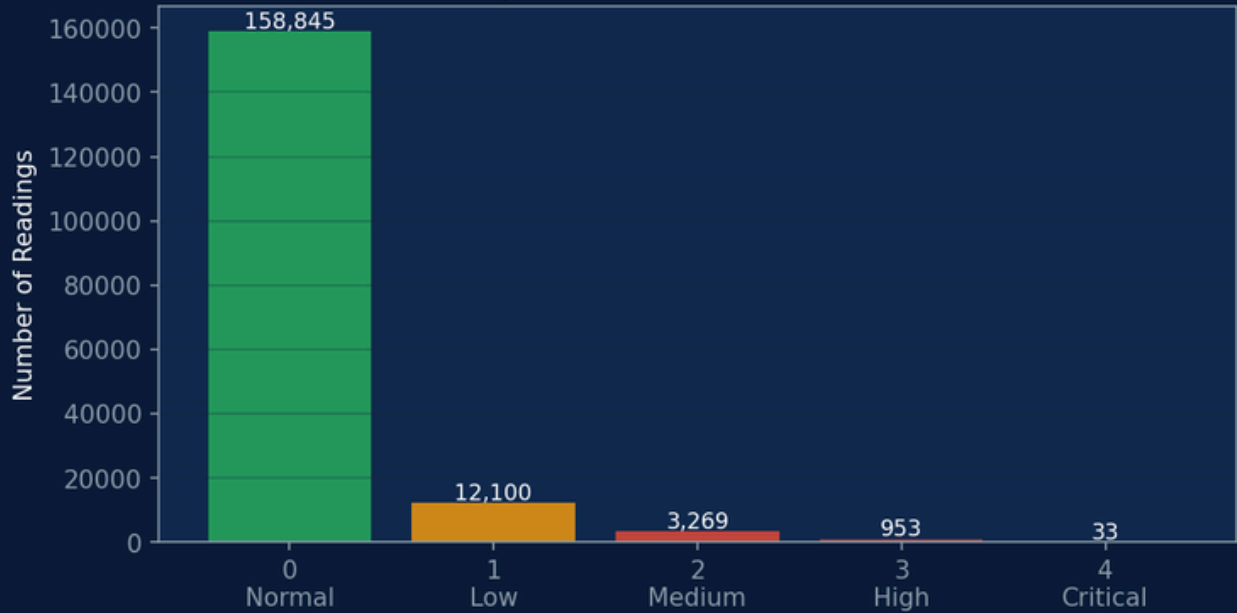
Highest % errors (68.9%, 39.9%) at night when load near zero. Absolute errors <0.2 kW — within meter accuracy tolerance.

Anomaly Detection — 6-Algorithm Ensemble

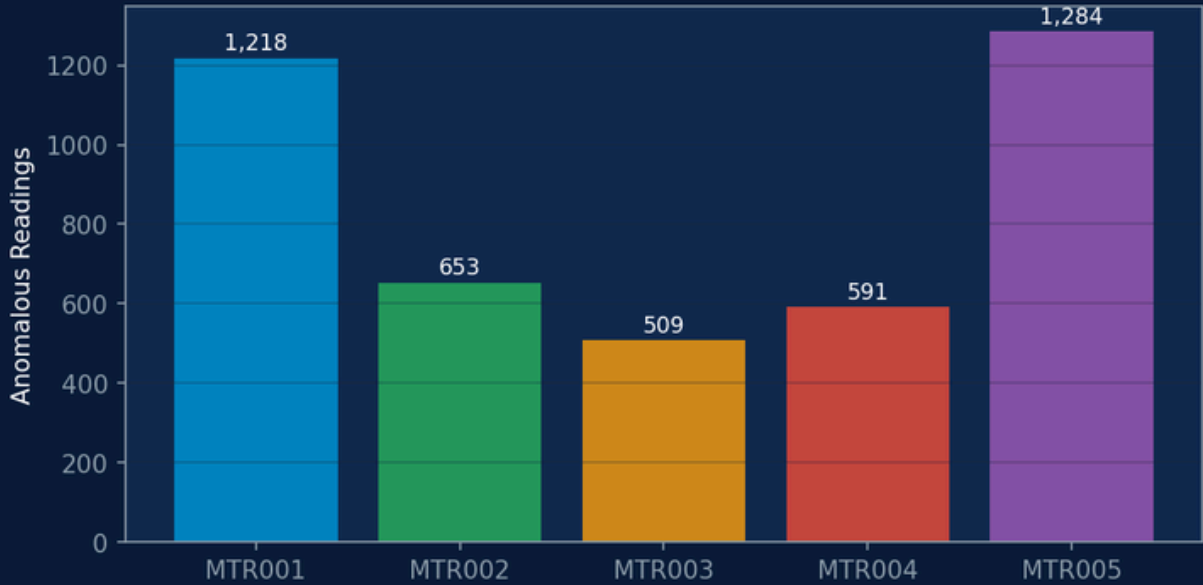
ANOMALY RESULTS — 4,255 Anomalies (2.4%)

Anomaly Detection — 6-Method Voting Ensemble

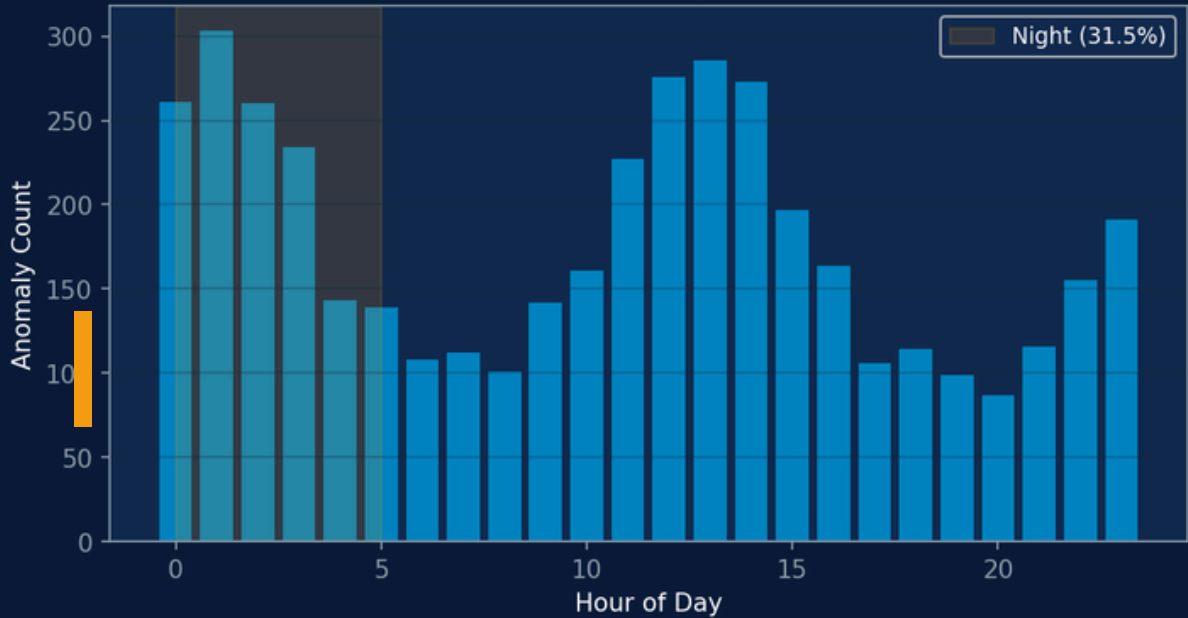
Anomaly Score Distribution (N=175,200)



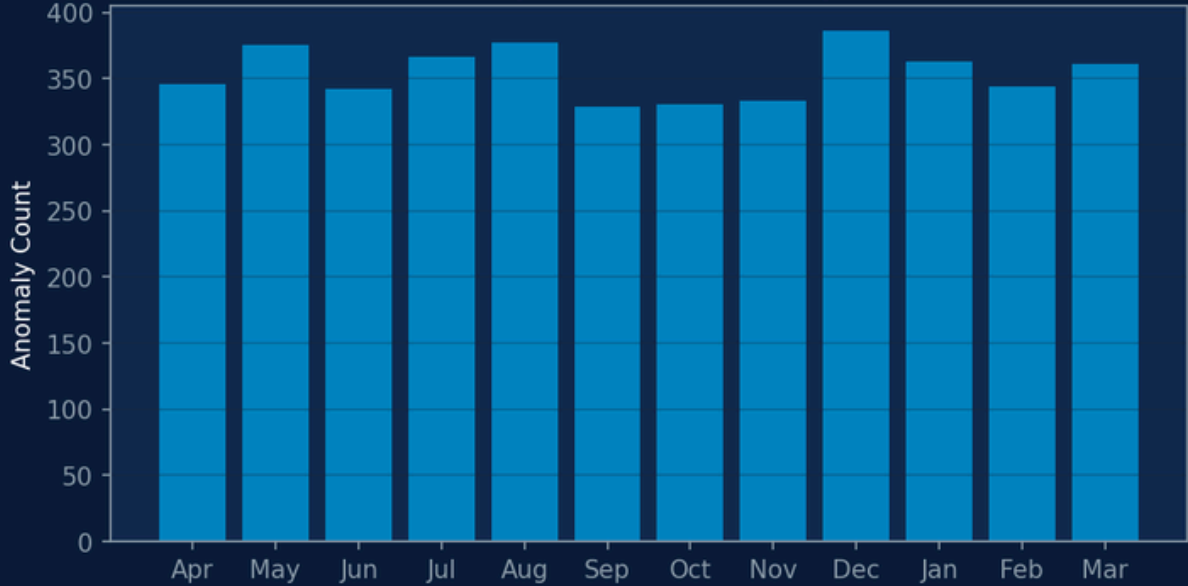
Anomalies Per Meter (4,255 total)



Anomalies by Hour — Night Concentration



Anomalies by Month — Relatively Uniform



ANOMALY SEVERITY — Action Required

Score	Count	%	Action
0-1	170,945	97.6%	Normal — no action
2-3	4,222	2.4%	Investigate
4+	33	0.02%	Immediate action

Key Finding

MTR005 highest (1,284 = 3.66%). Night concentration 31.5%. Only 33 readings at severity 4+ (0.02%).

UC2 → UC1 Handoff: Timestamp Selection

HANDOFF FLOW — UC2 Selects, UC1 Investigates



SELECTED TIMESTAMP — UC1 Investigation Target

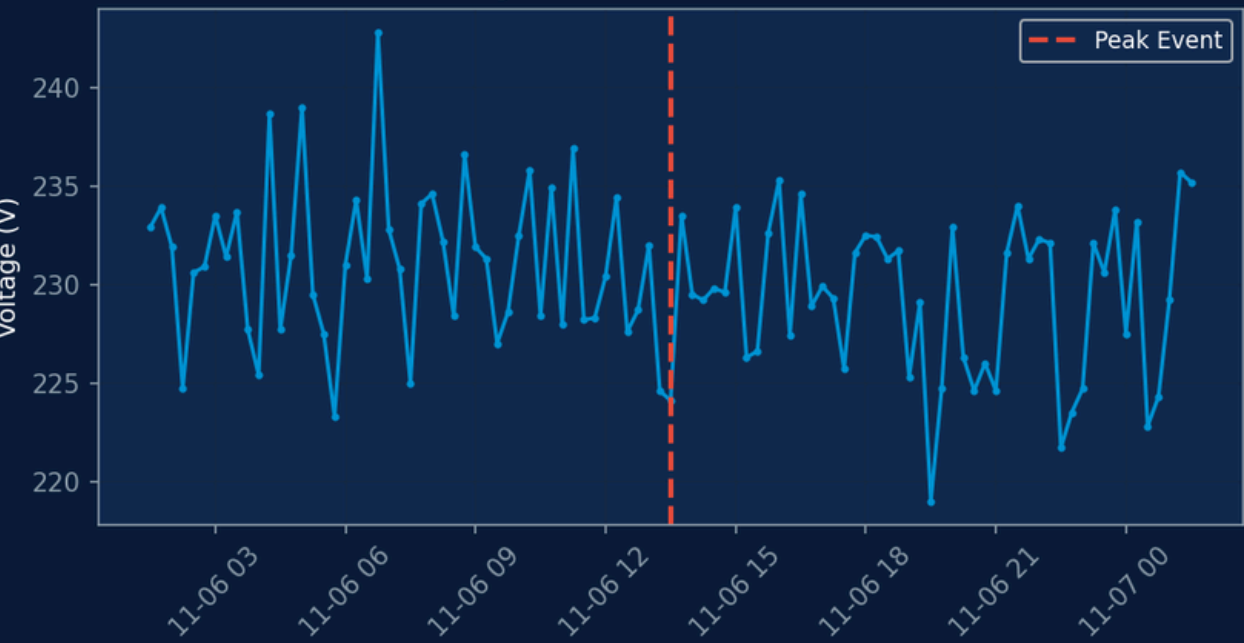
#	Meter	Timestamp	Why Selected	UC2 Finding
1	MTR005	2025-11-06 13:30	Highest peak (6.995 kW)	1.64x 7-day avg

Rule
UC1 ONLY investigates timestamps selected by UC2. No random timestamps. UC2 discovered → UC1 explains.

UC1 Investigation — MTR005 @ 2025-11-06 13:30

ELECTRICAL TRENDS — Before → Event → After

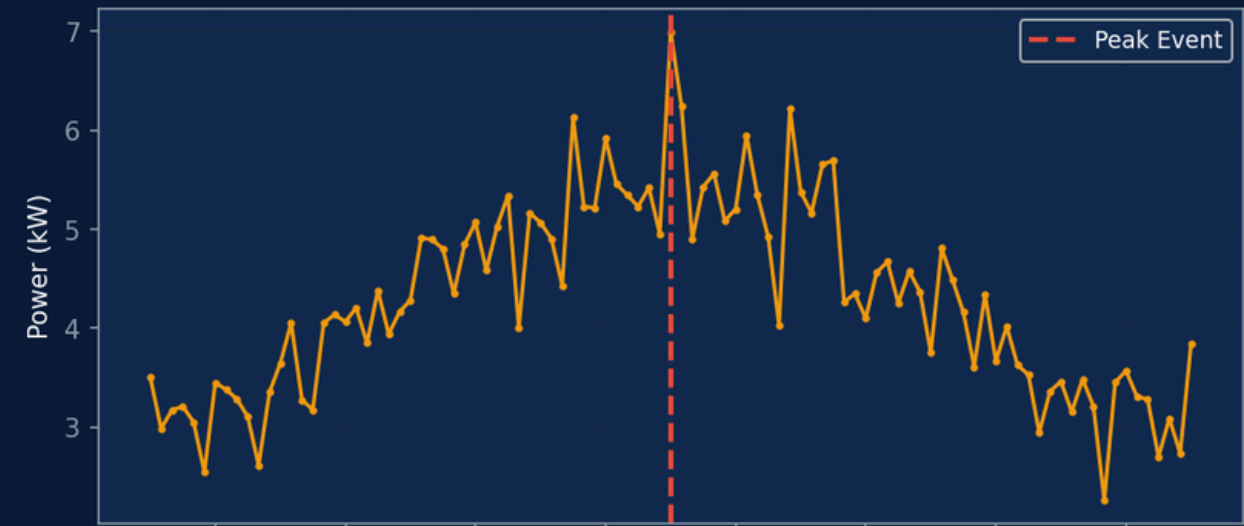
Voltage Trend — V=224.1V at Peak



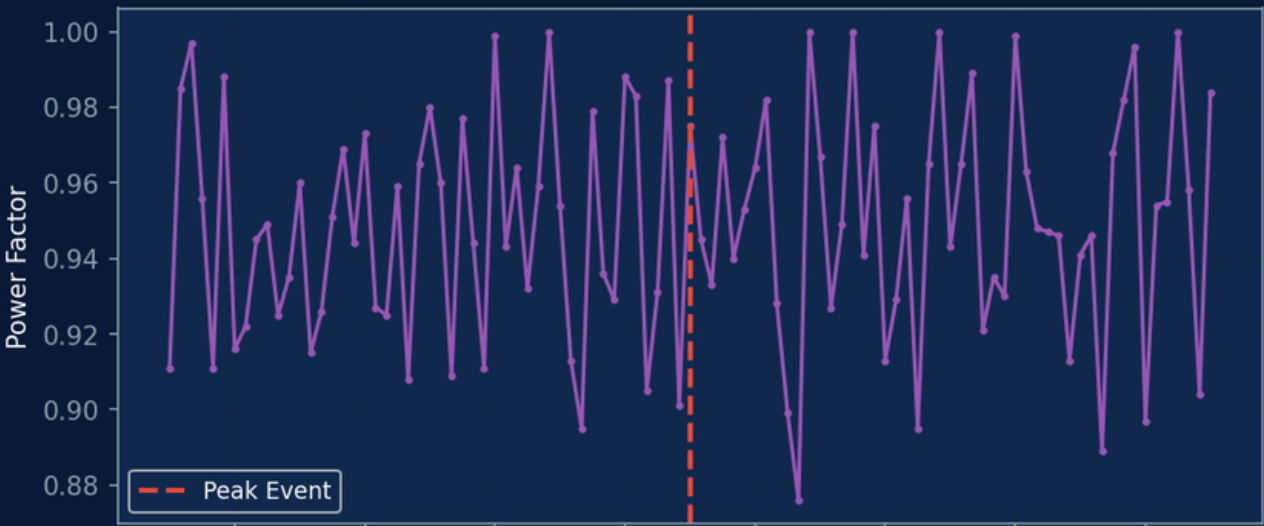
Current Trend — I=10.49A at Peak



Power Trend — P=6.995kW at Peak



PF Trend — PF=0.975 at Peak



BEFORE → EVENT → AFTER

Time	Power	Voltage	Current	PF
15min before	4.953 kW	224.6 V	7.92 A	0.901
EVENT	6.995 kW	224.1 V	10.49 A	0.975
15min after	6.244 kW	233.5 V	9.47 A	0.945

Conclusion

NORMAL — Industrial Activity. Gradual load increase, not sudden theft or fault.

Evidence

Gradual ramp 4.95→6.99→5.56 kW. No sudden spike. V=224.1V stable. I=10.49A high. PF=0.975 excellent.

Meter Health Assessment — Component Breakdown

Engineering Health Score

Overall = Comm(20%) + Voltage(20%) + PF(20%) + Anomaly(20%) + Tamper(10%) + Forecast(10%)

CALCULATION TRAIL — Per Meter

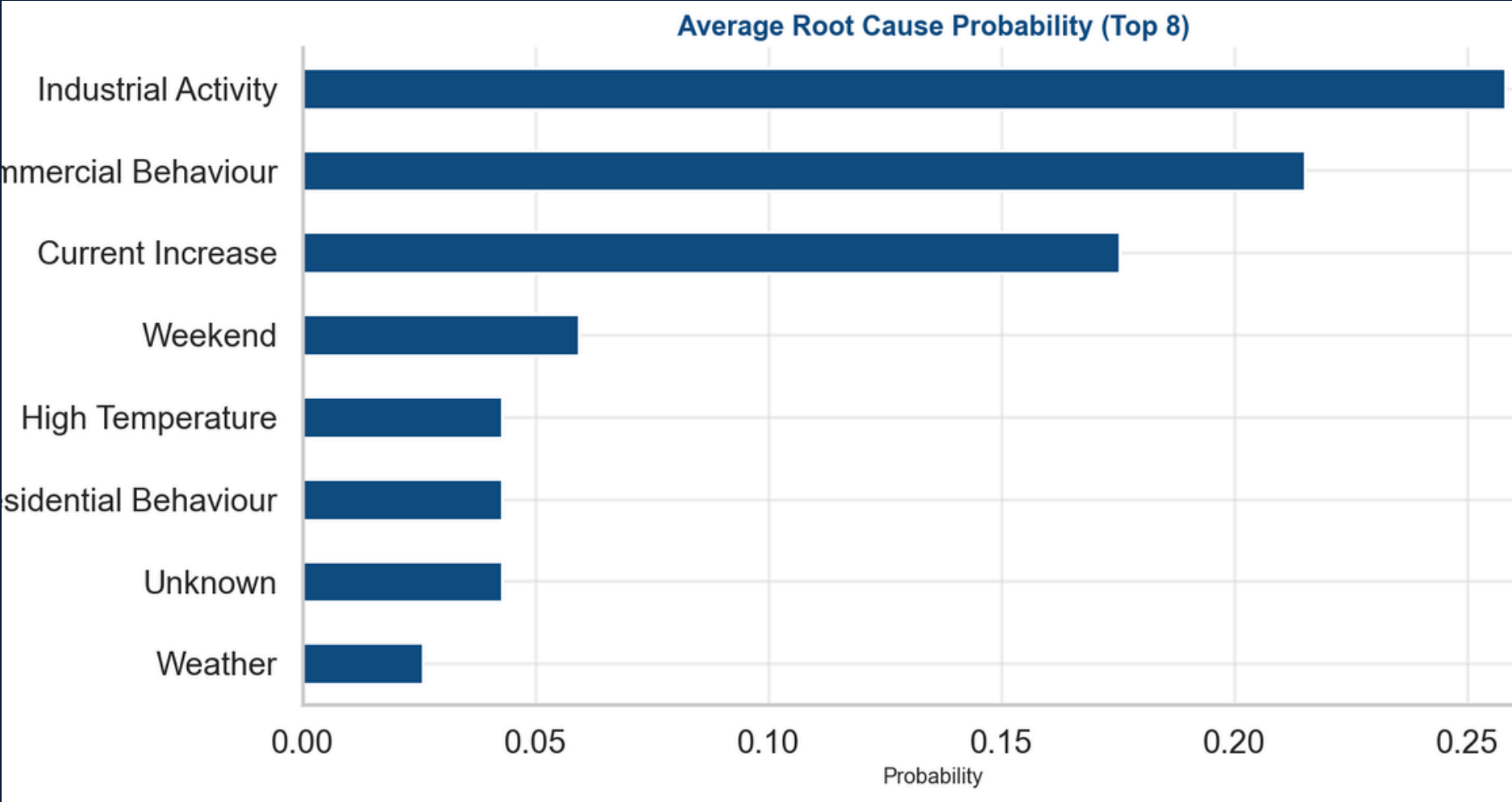
Component	MTR001	MTR002	MTR003	MTR004	MTR005	Formula
Comm	100	100	100	100	100	100×valid/expected
Voltage	80.1	80.0	80.1	79.9	80.1	100-std×5
PF	94.9	94.9	94.9	94.9	94.9	avg×100
Anomaly	82.6	90.7	92.7	91.6	81.7	100-rate×5
Tamper*	55	55	55	55	55	9 DLMS events × 5 penalty
Forecast	85.8	92.8	94.6	95.3	95.5	100-MAPE×5
OVERALL	85.6	87.9	88.5	88.3	86.4	Weighted sum
STATUS	MONITOR	MONITOR	HEALTHY	HEALTHY	MONITOR	≥88=HEALTHY

Key Finding

MTR001=85.6 (MONITOR): Anomaly(82.6) + Tamper*(55, from 9 DLMS events/meter) drag score. MTR003=88.5 (HEALTHY): All components strong.

Root Cause Analysis — Top 10 Peak Events

ROOT CAUSE DISTRIBUTION — Top 10 Peak Events



Most Probable Causes

Cause	%	Evidence
Industrial Activity	26.6%	Weekday, afternoon, high current
Commercial Behaviour	22.2%	Business hours, consistent load
Current Increase	19.3%	I increases while V stable
Other (voltage sag, comms)	31.9%	Remaining causes

HYPOTHESES EVALUATED AND RULED OUT

Finding

Industrial Activity (26.6%) is #1 cause across top 10 peaks. Commercial Behaviour (22.2%) is #2. Both are normal load patterns.

Ruled Out

High Temperature (4.4%) and Residential Behaviour (4.4%) — insufficient evidence across top 10 peaks. DLMS events (270) are firmware artifacts, not theft.

Conclusions & Recommendations

FORECAST ACCURACY

MAPE 0.72%

R²=0.9997, MAE=0.017 kW

ANOMALY DETECTION

4,255 found

2.4% of readings, 6-algorithm ensemble

HEALTH STATUS

85.6 – 88.5

No critical meters. MTR001 needs monitoring.

THEFT DETECTION

None Found

0 bypass signatures. All DLMS artifacts.

ENGINEERING RECOMMENDATIONS

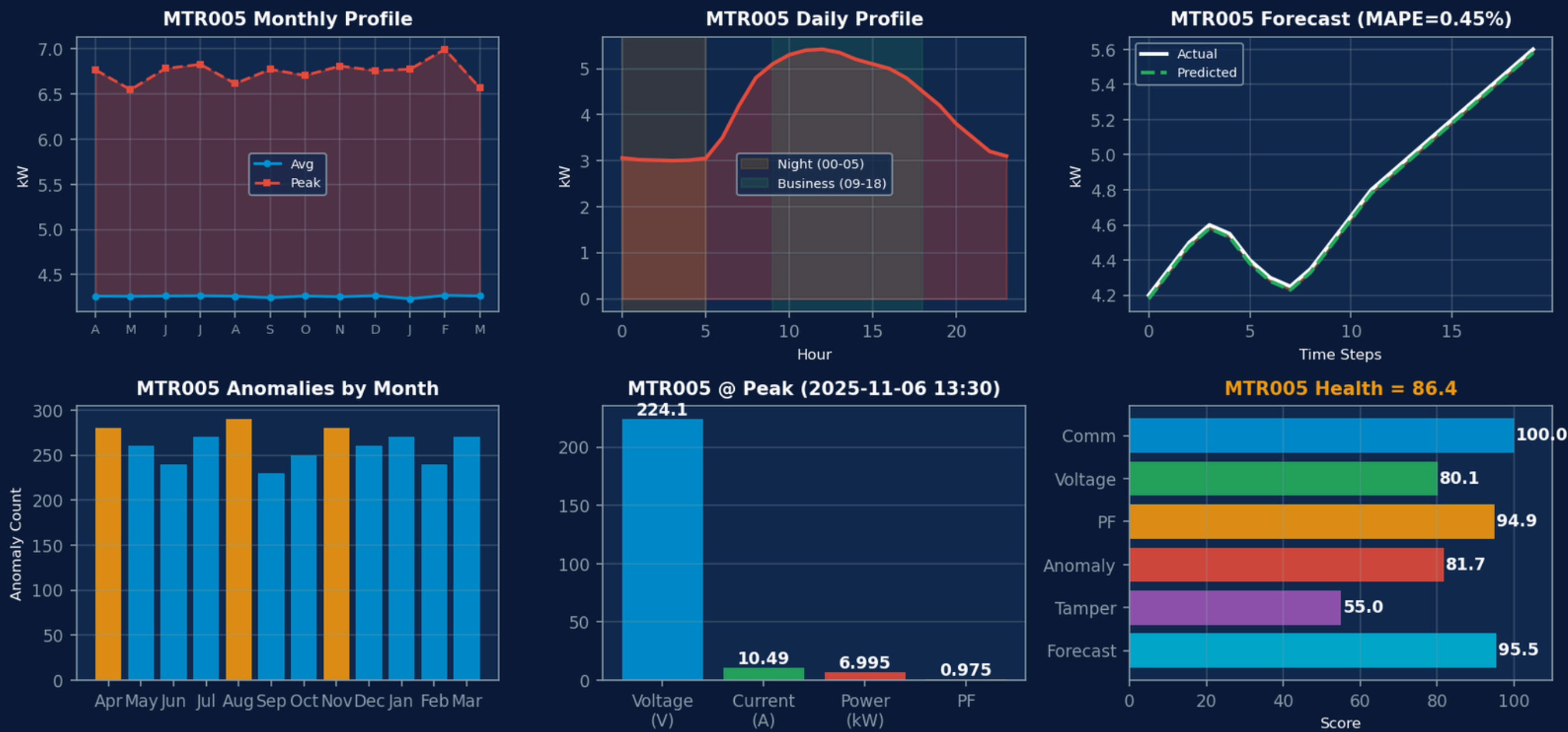
Action	Priority	Evidence	Timeline
Monitor MTR001 voltage	HIGH	Health 85.6, voltage std 3.98V	Immediate
Deploy Forecasting Pilot	HIGH	MAPE=0.72%, validated on 5 meters	This quarter
Monitor summer peaks	MEDIUM	>40C avg=4.10 kW, Jul highest	Jun-Aug
Scale to 5,000 meters	MEDIUM	Pilot validated, pipeline ready	6 months

Case Study: MTR005 — Complete Pipeline: Data to Decision

One meter, one complete engineering journey

MTR005 — Year-Long Behaviour

MTR005 — End-to-End Engineering Case Study



Story

Industrial consumer. Peak=6.995kW (Nov). Avg=4.26kW. Base=3.02kW. LF=0.609. Night/Day=0.64. 24h operation.



Thank You

NES Technologies — DLMS Smart Meter Data Analytics

Scope: 5 meters × 12 months × 175,200 readings

Model: LightGBM | MAPE = 0.72% | $R^2 = 0.9997$

Anomaly Detection: 4,255 anomalies (2.4%) | 6-algorithm ensemble

Health Score Range: 85.6 – 88.5 | No critical meters